5
Debt Management

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1. Introduction

Sound public debt management is a key element in ensuring debt sustainability. Debt structures that are robust to interest rate, exchange rate, or other shocks potentially allow countries to sustain higher debt levels at times of difficulty. Sound public debt management practices help them to finance their borrowing requirements and lessen risks embedded in their public debt portfolios, reducing the risks to government balance sheets and to wider financial stability (Jonasson and Papaioannou 2018). The identification of possible funding sources for financing needs, along with the identification, measurement, and management of public debt portfolio risks, are critical in determining appropriate debt strategies.

In assessing and deciding debt strategies, a debt manager traditionally analyzes a broad set of issuance strategies (e.g., foreign- vs. domestic-currency debt, fixed- vs. floating-rate debt, short- vs. long-term maturities). These strategies are designed to finance a pre-defined fiscal deficit, but are typically constrained by the country's debt risk indicators (IMF-WB 2014). In essence, each debt issuance strategy is assessed on the basis of its implied debt service costs and its impact on the relevant risk indicators. Stress tests determine the impact of changes in risk factors (e.g., exchange rates, interest rates) and the underlying macroeconomic conditions, on relative debt costs and risk indicators. Based on the robustness of these strategies to risk shocks, the debt manager ranks the different strategies.

For most middle-income and low-income countries, balancing cost and risk often remains an aspirational objective. Many developing countries have limited access to capital markets, domestic and external, and debt managers do not have the opportunity to choose from an array of options.

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Instead, they are usually faced with limited, and relatively risky, sources of financing. Consequently, a debt manager in a developing economy will often be preoccupied with seeking to develop debt markets, to establish new financing options. Objectives are broader and could include: (i) building a yield curve facilitating the private sector's pricing of risk; (ii) extending public debt maturities; (iii) creating benchmark issuances and building liquidity of secondary markets; (iv) diversifying the country's investor base, by attracting foreign investors or a new investor class, or developing new financial products for existing markets (e.g., inflation-linked bonds, zero coupon bonds); (v) supporting financial sector development, and in general promoting financial stability; and (vi) building sufficient foreign exchange reserves to weather a crisis (either from within or abroad).

The chapter is organized as follows: Section 2 presents the main debt management objectives from (i) a portfolio perspective and (ii) a wider policy perspective; Section 3 outlines the drivers of risk in debt structures, and the related issues of contingent liability management and debt sustainability; Section 4 presents some domestic and external debt issues, including the concept of "original sin" and an assessment of the investor base for advanced economies; Section 5 provides an overview of critical aspects of financial markets for debt management; Section 6 touches on some institutional and governance issues, including the interplay between cash and debt management, and the role of debt as a safe asset for the financial sector; Section 7 concludes with some comments on the lessons from recent experience.

2. Debt Management Objectives

A. From a portfolio perspective

The main objective of public debt management is to ensure that the government's financing needs and its payment obligations are met at the lowest possible cost over the medium to long run, consistent with a prudent degree of risk. Prudent risk management to avoid risky debt structures and strategies (including monetary financing of the government's debt) is crucial. The macroeconomic consequences of a public debt default and the magnitude of the ensuing output losses can be especially severe (see Chapter 7). These costs include business and banking insolvencies as well as the diminished long-term credibility and capability of the government to mobilize domestic and foreign savings.

Minimizing cost, while ignoring risk, should not be an objective. Transactions that appear to lower debt servicing costs often embody significant risks for the government and can limit its capacity to repay lenders. Managing cost and risk therefore involves a trade-off. Judgments will have to be made based on the risk tolerance of the government, keeping in view other policy objectives and policy buffers. The conceptual approach underpinning the preparation of a debt management strategy is outlined in Case Study 5.1.

Case Study 5.1 Developing a medium-term debt management strategy

The medium-term debt management strategy (MTDS) operationalizes a country authorities' debt management objectives; it is a "plan that the government intends to implement over the medium term (typically 3–5 years) in order to achieve a desired composition of the government debt portfolio, which captures the government's preferences with regard to the cost–risk tradeoff" (IMF and World Bank 2009). In this context, "cost" is the cost of servicing the debt, usually expressed as a ratio of GDP; and "risk" is the volatility of debt servicing costs. The basic methodology explores how different issuance strategies perform against a range of macroeconomic scenarios.

Conceptually the task is to identify efficient issuance mixes. This can be expressed in the form of a cost–risk boundary (or indifference curve) where policymakers can choose their preferred trade-off between cost and risk. This choice is essentially a political one, because of the implications it has for inter-generational equality–see Figure 5.1. As Herbert Hoover said, "blessed are the young because they will inherit the national debt."

The same principles apply to assets as well as liabilities, and more developed countries try to integrate the management of assets and liabilities by matching the cash flows, at least of sub-portfolios (for example borrowing to finance a project and on-lending to the relevant executing agency on the same terms and conditions). In that way, they are better hedged against shocks. Insofar as this matching is possible across most assets and liabilities, the framework boils down to matching the present value (PV) of the fiscal position (i.e., the PV of future taxes less the PV of future expenditures) with the PV of debt, which is the equivalent of deferred taxes. On this approach, the ideal debt structure generates servicing costs positively

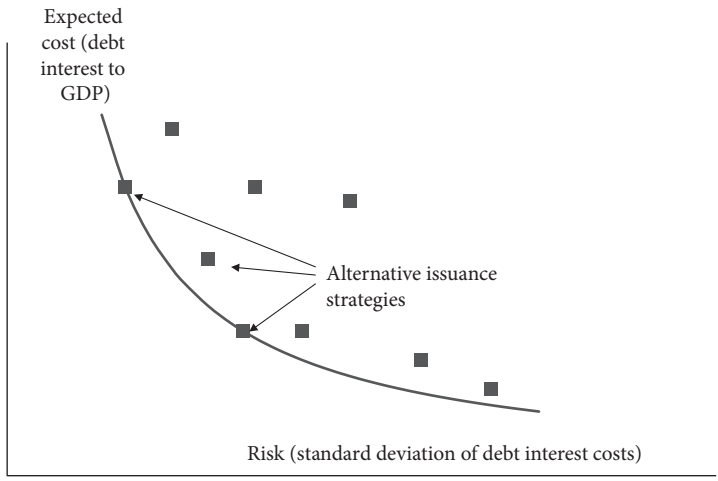

Figure 5.1 The cost–risk trade-off

linked to government revenues, sometimes referred to as "fiscal insurance." If the impact of adverse economic shocks can be minimized in this way, it reduces the risk of having to increase taxes and the losses that that imposes on the economy. Similarly, countries that are susceptible to demand shocks may choose to issue index-linked debt (because revenues and servicing costs will tend to move together); but if there are greater risks from supply shocks that might be exactly the wrong strategy (see Chapter 9).

Cost-at-Risk (CaR) techniques can be used to rank different issuance strategies and liability management operations on the basis of their performance under different shocks. The ranking is determined within an optimization framework, where an objective function (the debt costs and risk indicators) and constraints (the available instruments and the prevailing macroeconomic framework) are explicitly specified by the debt manager. Stochastic simulations of scenarios allow each debt strategy or instrument mix to be assessed on the basis of how relative debt costs and risk indicators respond. The simulations assess the robustness of the different debt strategies by showing the probability of debt servicing costs exceeding a specified threshold (the CaR), and the strategies are ranked accordingly. It should be noted that the CaR approach does not take into account refinancing and credit risks, and that the implied debt issuance strategies and/or liability management operations may not always be feasible.

The CaR approach does allow an assessment of debt costs with a probability structure. Monte Carlo simulation methods are used to derive a

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Case Study 5.1 *Continued*

cost–risk efficient frontier for portfolio structure or debt composition; on the efficient frontier, the cost of any debt composition is the lowest for any chosen risk level. The set of all such compositions reflects the cost–risk trade-offs that are faced by the debt manager. Given the government's choice of an acceptable level of risk (or the maximum acceptable level of cost, which may in practice be determined by debt sustainability considerations), it is possible to choose a specific optimal debt portfolio on the frontier, that is, the benchmark portfolio. The target benchmark portfolio can then be used to design a country's actual issuance schedule and liability management operations.

Developed economies, which typically have deep and liquid markets for government securities, often focus primarily on market risk, and, together with stress tests, may use sophisticated portfolio models for quantifying and measuring this risk. In contrast, emerging market and low-income economies, which may have only limited (if any) access to foreign capital markets and also relatively undeveloped domestic debt markets, should give higher priority to refinancing risk, that is, the risk of much higher interest rates or of difficulty accessing the market when it is time to refinance a bond. Where appropriate, policies to promote the development of the domestic debt market should be included as a prominent government objective. This objective is particularly relevant for countries where market constraints are such that short-term, floating rate, and foreign currency debt may, in the short run at least, be the only viable alternatives to monetary financing.

Poorly structured debt portfolios, in terms of maturity, currency, or interest rate composition and large contingent liabilities, have been important factors in inducing or propagating economic crises. For example, irrespective of the exchange rate regime, or the debt currency involved, crises have often arisen because of an excessive focus by governments on possible cost savings associated with short-term or floating rate debt (see Case Study 5.2 below for Mexico's 1994 experience). Issuance of large volumes of such debt instruments has left government budgets seriously exposed to changing growth and financial market conditions, including changes in the country's creditworthiness, when this debt has to be refinanced.

Excessive reliance on foreign currency debt poses particular risks, as it can lead to exchange rate and/or monetary pressures if investors become reluctant

to refinance the government's debt. This reluctance may be driven by external events; for example, many countries found external sources of finance drying up or becoming more expensive following the East Asian crisis of the 1990s. By reducing the risk that the government's own debt portfolio will become a source of instability for the private sector, prudent government debt management, along with sound policies for managing contingent liabilities (discussed further in Section 3.B), can make countries less susceptible to contagion and financial risk. Further, a debt portfolio that is robust to shocks leaves the government better placed to manage financial crises.

B. From a wider policy perspective

Sound debt management strategies can be instrumental in supporting financial stability, by creating a liability structure for public debt that sustains low refinancing risk for the sovereign throughout the business cycle. Debt management policies also interact with other government policies. Financial institutions typically hold a significant share of public debt in most countries. It is primarily the responsibility of the regulatory authorities to set and monitor financial institutions' prudential standards. But debt managers must recognize that their actions can have a very major impact on these institutions' balance sheets. Moreover, given the usually high level of interdependence of financial institutions, the effects can potentially have systemic implications.

This impact is relevant not only when discussing possible sovereign liability management and debt restructuring operations, but also when assessing the targeted debt composition. For example, although short-term debt involves higher refinancing risk, which could pose a higher risk to financial stability, longer-term debt may leave the holders' assets more exposed to the impact of interest rate fluctuations on their market value, that is, they represent higher value at risk (VaR) for the debt holder. Banking crises in Turkey and Uruguay in the 1990s and early 2000s were made far worse by the banks' exposure to longer-dated government bonds. Fixed rate bonds pose less risk to the government but may represent a higher risk to the investor. If individual investors, in search of higher profits, increase their exposure to interest rate risk and there is a hike in interest rates, the market as a whole may suffer as the unwinding of positions by some institutions triggers VaR thresholds for others.

Monetary policy raises similar issues. The long-established drive to separate fiscal and debt management policies on the one hand and monetary policy operations on the other, is underpinned by a recognition, at least in more advanced countries, of the weak links between debt management choices and

monetary conditions. This model, however, tends to assume highly liquid markets, and independence of policy instruments, which is not the case in less developed markets, where cash and debt operations may affect monetary policy operations and vice versa. Moreover, in developed markets this model is under strain from quantitative easing whereby central banks have set out to reduce or hold down longer-term interest rates, potentially affecting debt managers' decision making (Turner 2014). In all markets there needs to be some mechanism to facilitate high-level policy coherence. Arguably the government's portfolio strategy should take into account not only the structure of its own assets and liabilities, but also those of the central bank (both its government debt holdings and foreign currency reserves) (see Togo 2007).

Debt managers should be aware of and try to monitor their impact on these wider risks. They require a degree of coordination, or at least an opportunity for consultation, at the policy level, without jeopardizing the formal separation of operational decision making and the benefits in terms of clarity, focus, and accountability that flow from that.

3. Key Risks to Manage

A. Debt structures

In addition to ensuring that the government's financing needs and financial obligations over the medium- to long-term are met at the lowest cost consistent with a prudent level of risk, debt managers should target a sustainable debt service profile consistent with the government's medium-term debt repayment capacity and identify, measure, and manage debt portfolio risks (see OECD 2015).

There are clear trade-offs: on the one hand the superficial attraction of lower costs and a lower deficit in the short run, even if that means greater risk exposure; on the other hand, governments' responsibility to avoid exposing their portfolios to risks of large or catastrophic losses, even if they carry low probabilities. There are different drivers:

- *Maturity structure.* A government faces an inter-temporal trade-off between short-term and long-term costs that should be managed prudently. For example, excessive reliance on short-term or floating rate debt to take advantage of lower short-term interest rates may leave a government vulnerable to volatile and possibly increasing debt service costs, and even the risk of default if it cannot refinance its debts at any cost.

- *Foreign exchange exposure*. This can take many forms, but the predominant one is directly issuing excessive amounts of foreign-currency denominated debt and foreign-exchange indexed debt without hedging. This practice may leave a government vulnerable to volatile and possibly increasing debt service costs if its exchange rate depreciates, and again the risk of default.
- *Debt with embedded put options* This involves the ability of the holder to require redemption before maturity. If poorly managed, such options increase uncertainty to the issuer, effectively shortening the portfolio duration, and creating greater exposure to market/rollover risk.
- *Debt with early cancellation clauses*. Early termination events or rating trigger clauses can pose risks for debt management and need proper consideration.
- *Derivatives* (other than plain vanilla swaps aimed at reducing risk or hedging against movements in interest or exchange rates). Swaptions are sometimes used to alter current financial costs at the expense of higher future volatility.

The way that debt structures interact with market conditions can be particularly problematical. In addition to non-concessional domestic sources of financing, a number of emerging economies, in sub-Saharan Africa and elsewhere, are accessing international capital markets. Eurobond issuance has surged during a period of prolonged low global interest rates. However, the Eurobond market can be volatile and access conditions highly uncertain. Global interest rates are expected to increase and capital flow reversals could coincide with the initial wave of Eurobonds reaching maturity in the early 2020s. Refinancing risk could become acute, particularly for countries with macroeconomic imbalances.

B. Contingent liabilities

Risks do not arise only from the interaction between debt structures and economic shocks. Governments are also exposed to risks arising from both contingent liabilities—often in the form of a guarantee, that is, the promise to service the debt of a beneficiary if it fails to do so—and assets whose servicing is contingent, usually stemming from on-lending by government to a beneficiary. This credit risk exposure flows from the beneficiary's inability or unwillingness to service its debt, whether to a third party (in the case of a

guarantee) or government (in the case of on-lending). As discussed in Chapter 2, there are other contingent liabilities, both explicit and implicit; they include the potential costs to the central government flowing from public private partnerships (PPPs), the debt of state-owned enterprises (SOEs) or lower levels of government, (natural) disasters, and financial sector crises. A sound and prudent risk management framework includes well-defined risk management objectives, an analysis of risks, and the design and implementation of a risk management strategy incorporating monitoring, reporting, and reassessment procedures.

Requests for guarantees must be individually assessed and all contingent liabilities individually identified, analyzed, and monitored; but the aggregate exposure also needs to be assessed in the context of the government's balance sheet structure. Even though contingent liabilities are not recorded on the balance sheet, they can still pose a huge threat to government finances and should be included alongside other sovereign exposures, in order to systematically identify the interaction between the different risks and how they might be differentially affected by different shocks. Debt sustainability analysis (DSA) should normally include, as one of the potential shocks, how the crystallization of contingent liabilities might affect debt-to-GDP ratios or the path of debt service over time. But the debt management strategy analysis should take this a step further by considering how the balance sheet, as a whole, might be affected by different shocks and the scope for hedging the residual risks.

Sovereign debt managers use different risk management tools for contingent liabilities. Some operate at the level of the individual proposals, with an assessment of the risk and budget exposure, measured against the benefits. Credit risk fees should be charged for the guarantee, both as a way of mitigating the government's risk and ensuring that the project is properly assessed. The fees may flow to the budget or be retained in a fund (which may have earmarked assets or be retained in the general ledger) to meet the cost of guarantees being called. Initially, such fees may not be based on thorough credit risk analysis, but be applied at a flat rate. Over time, however, risk managers ideally refine risk mitigation and management tools and differentiate fees based on the projects' or beneficiaries' credit quality.

Similarly, there are different techniques for managing the aggregate risk from contingent liabilities. The crude but simple—and commonly applied—approach is to set a limit on the issue of guarantees, reflecting risk appetite. Several countries have adopted a fiscal target or indicator that is defined in terms of central government debt plus government-guaranteed debt. Although

the contingent liabilities are not recorded as debt unless and until they crystallize, the aggregate is reported alongside debt data. As sophistication grows, so do the tools. The limit might be set on expected exposure, rather than on the nominal value of the guarantee. Some countries have built econometric models that identify the expected losses in any one year and include provisions for losses in the budget accordingly.

C. Debt sustainability

Debt sustainability is analytically different from debt management. But the risk with any debt composition may depend on the level of debt; and countries with, say, a high debt-to-GDP ratio need to be especially careful about their debt composition and exposure to shocks. The Mexican tequila crisis of 1994–95, summarized in Case Study 5.2, is an example of how this interaction can have damaging effects.

Case Study 5.2 Debt management under sovereign stress: Mexico's Tequila crisis (1994–95)

The 1994 Mexican peso crisis, a currency crisis sparked by the Mexican government's sudden devaluation of the peso against the US dollar in December 1994, ignited a capital flight that became an international financial crisis, known also as the "Tequila crisis."

During the 1994 presidential election, the incumbent administration embarked on an expansionary fiscal and monetary policy. Mexico's current account deficit grew to about 7 percent of GDP, and Mexico's treasury started issuing short-term peso-denominated treasury bills with a guaranteed repayment denominated in US dollars, called "tesobonos." The tesobonos offered a lower yield than Mexico's traditional peso-denominated treasury bills, called "cetes," but their dollar-denominated returns were more attractive to foreign investors. Mexico enjoyed investor confidence and new access to international capital following the entry into force of the North American Free Trade Agreement (NAFTA) in January 1994. However, domestic unrest and political instability increased investors' risk premium on Mexican financial assets.

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Case Study 5.2 *Continued*

The higher risk-premia initially had no effect on the peso's value because Mexico had a fixed exchange rate. The central bank intervened in the foreign exchange markets to maintain the peso's peg to the US dollar by issuing short-term dollar-denominated public debt to buy pesos. With the peso growing stronger, domestic businesses and consumers purchased more imports, and Mexico began running a large trade deficit. Speculators' increasing recognition of the peso's overvaluation contributed to capital flight, further reinforcing downward pressure on the peso. Under election pressures, Mexico purchased its own treasury securities to maintain its money supply and avert rising interest rates, drawing down the bank's dollar reserves. Supporting the money supply by buying more dollar-denominated debt, while simultaneously honoring such debt, that is, servicing the tesobonos with US dollar repayments, depleted the bank's reserves.

On December 20, 1994, newly inaugurated President Ernesto Zedillo announced a devaluation of the peso, and foreign investors' fear of additional devaluations generated an even higher risk premium on domestic assets. This increase placed additional upward market pressure on Mexican interest rates as well as downward market pressure on the Mexican peso. To discourage the resulting capital flight, the central bank raised interest rates, but higher borrowing costs merely hurt economic growth prospects. When the time came for Mexico to roll over its maturing debt, the government was unable to sell new issues. Further, to repay tesobonos, the central bank had little choice but to purchase dollars with severely devalued pesos, which proved extremely expensive. Under these circumstances, the government faced an imminent sovereign default.

On December 22, the bank allowed the peso to float freely, after which it continued to depreciate significantly. The Mexican economy experienced hyperinflation of around 52 percent (by end 1995) and mutual funds began liquidating Mexican assets, as well as emerging market assets in general. The crisis led to financial contagion throughout developing countries, spreading quickly to economies in Asia and the rest of Latin America. The United States organized a US$50 billion bailout for Mexico in January 1995, administered by the IMF with the support of the G7 and Bank for International Settlements. In the aftermath of the crisis, several of Mexico's banks collapsed amidst widespread mortgage defaults. The Mexican economy experienced a severe recession and deteriorating poverty and unemployment.

As discussed in Chapter 4, debt sustainability encompasses the concepts of solvency and liquidity. There is a range of indicators, although key ones include the debt-to-GDP ratio and debt servicing requirements over time, compared, for example, to expected government revenue. Debt management and debt sustainability have a different focus. The DSA focuses on medium-term vulnerabilities; it primarily feeds into fiscal policy decisions, notably on the primary balance. Debt management strategy analysis, on the other hand, focuses on the composition of the debt portfolio and its inherent risks, rather than its overall size.

The interaction goes both ways. The DSA recognizes that vulnerabilities can arise from the composition of the debt portfolio. But it is not designed as a tool to analyze how debt compositions can be altered to reduce those vulnerabilities and whether this is cost effective: that is the task of debt management. There are several ways in which the risk with any debt composition may depend on the level of debt. They include home country bias among investors, discussed further in Case Study 5.3. Others include the impact of shocks on domestic markets, the effectiveness of fiscal policy and the country's net asset position, as well as more obviously political and governance issues. In practice debt managers and DSA analysts should work together, drawing on the same underlying data and using similar scenarios.

4. Domestic and External Debt

A. The constraints on emerging market economies

Borrowing choices are constrained; and borrowing costs may depend on what is or has been practical. Emerging Market (EM) governments often face limited appetite for their securities in the local currency market, and may prefer to tap broader and more liquid markets in the major international currencies. In the 2010s, as in the EM bond issues of the 1980s and 1990s, many international bonds were issued in foreign currencies (mostly US dollars). The domestic currency markets were (and frequently still are) too thin and shallow, or virtually absent, in particular for long-term maturities. Issuers may also have had some opportunistic reasons and attempted to lower the cost of servicing their debt by exploiting lower foreign currency interest rates (Abbas et al. 2014).

Borrowing in foreign currencies can also be a result of "original sin." These sovereigns might be willing but unable to borrow in local currency in the international markets as a result of investors' lack of trust in the sovereign,

based on their past transgressions—their "original sin." This problem (described in the seminal work by Eichengreen and Hausmann 1999), leads over time to external debt accumulation and currency mismatches on the balance sheet, exposing it to a real exchange rate depreciation. Another explanation for a country's difficulty in borrowing in its own currency is the limited benefit to investors' portfolio diversification objectives. The established practice in the international financial centers is to operate in a limited number of major currencies. As a result, developing countries, as latecomers to the international financial game, face an uphill battle when attempting to add their currencies to the international portfolio.

B. Advanced economies—investor base risk index

It is not news that EMs can be vulnerable to bouts of market volatility. Investors often trigger sudden stops—they stop buying or start selling off their holdings of government bonds. But it become apparent in the 2010s that advanced economy government bond markets can also experience investor outflows, and associated runs. At the same time, some safe haven countries have seen their borrowing costs drop to historic lows as they experience rising inflows from foreign investors, sometimes despite their domestic policies.

Several strands of research show that advanced economies' sovereign borrowing costs also depend on who is holding the bonds—the demand side for government debt. Regardless of their level of debt, some countries are more at risk of sudden investor outflows, and associated spikes in government borrowing costs, as a result of the risk characteristics of their investor base. Tracking who owns what, when, and for how long, can shed some light on potential risks in advanced economies' government debt markets. Standard measures of sovereign risk, such as debt-to-GDP ratios, are relevant; but it is important also to understand better the investor base for government debt, and why and how investors change their allocations.

In that context, Arslanalp and Tsuda (2014a, 2014b) have developed an "investor base risk index" that focuses on the stability of investor demand. The index runs from zero to 100, based on the composition of the investor base and the risk scores assigned to different investors, reflecting the way they tend to change their holdings. By this metric, countries with a high share of domestic investors, such as domestic banks and central banks, as well as foreign central banks, in their investor base receive lower scores. In contrast, high scores are assigned to countries whose investor base has a high share of

		Demand-side risk indicator	
		Low	High
Supply-side risk indicator	High	**Quadrant I** High debt but resilient to a run	**Quadrant II** High debt and prone to a run
	Low	**Quadrant III** Low debt and resilient to a run	**Quadrant IV** Low debt but prone to a run

Figure 5.2 A stylized framework for sovereign risk analysis

Note: Supply-side risk indicators could include gross debt-to-GDP ratio, net debt-to-GDP ratio, projected debt-to-GDP ratio or other measures of the supply of government debt. Demand-side risk indicators could include the investor base risk index constructed by the authors or any other measure that captures the refinancing risks inherent in the sovereign investor base.

foreign private investors, as empirically they are the most skittish in times of trouble.

The index is especially useful when used in combination with a supply-side risk indicator of government debt to assess the overall sovereign risk of a country. Figure 5.2 summarizes the thinking in a two-by-two table. The best of all worlds is represented by Quadrant III (low debt and low investor base risk), while the worst is represented by Quadrant II (high debt and high investor base risk).

Arslanalp and Tsuda (2014a) tested this idea by looking at how countries fared in terms of investor base risk at end-2009 before sovereign risk emerged as an issue for some advanced economies in 2010. The risk index suggests that several euro area countries were prone to a sudden stop as early as end 2009, before the later euro area debt problems fully came to light (Figure 5.3, Quadrant II). At the same time, countries with high debt and low investor base risk, which include Germany, Japan, and the United States, did not face similar market pressures despite their high projected levels of debt; while countries with low debt and low investor base risk, which included Australia, Canada, and Sweden, became the "new safe haven" countries during this period. Although hindsight is perfect, this classification is remarkably close to how markets ended up differentiating these countries in terms of sovereign risk after 2009.

This analysis also helps to explain the high debt, low yield puzzle. Both supply and demand dynamics drive trends on sovereign bond yields—and are likely to continue to do so in the future. Trends in the supply of government debt are relatively straightforward to follow and interpret, and

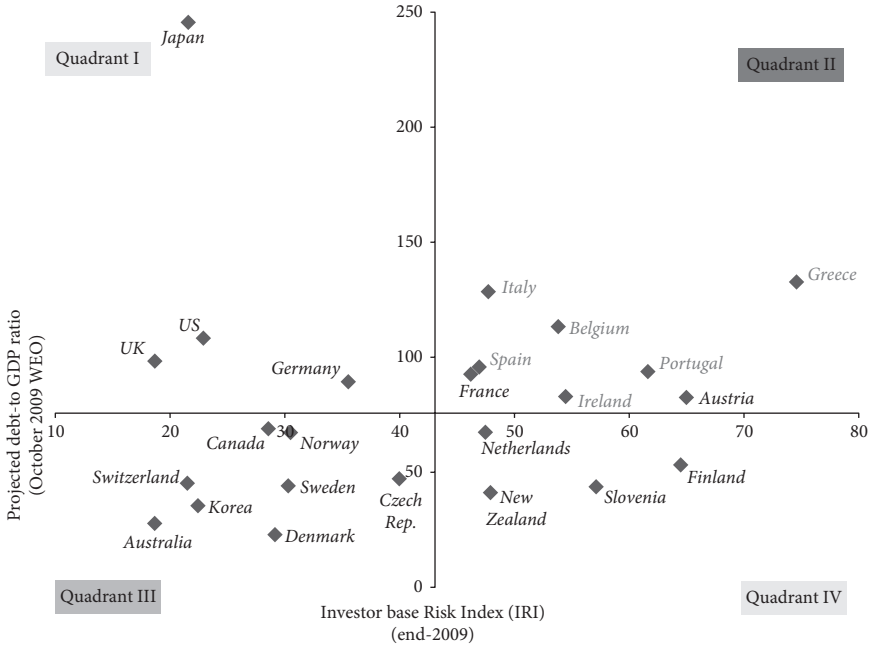

Figure 5.3 Advanced *Economies—Application of the Investor Base Risk Index*, end-2009

projected debt-to-GDP ratios are often available in, for example, the IMF's World Economic Outlook. But demand factors are more difficult to track and anticipate. Moreover, supply factors alone do not explain well recent changes in government bond yields. For instance, highly indebted governments like Japan still experience some of the lowest government bond yields among advanced economies.

The Investor Base Index can explain why some countries are able to sustain much higher levels of debt without pressure from financial markets. For example, despite continued worries about Japan's fiscal outlook, demand for government bonds, in particular from domestic investors, has been strong and bond yields have been stable—a fact captured by the risk index. This suggests that, where the investor base risk is low, high debt-to-GDP levels may matter less, as the likelihood of a run by sovereign investors is also lower. In that respect, the index suggests that Germany, the United Kingdom, and the United States may be in the same category as Japan, although for a different reason: those countries rank low in the index mainly because of the high share of foreign central banks in their investor base. Case Study 5.3 elaborates on the home country bias apparent in Japan.

Case Study 5.3 Home country bias in Japan

By the late 1920s Japanese companies had proved to be active issuers of corporate bonds. The government only reorganized its financial system into an indirect finance system—a bank-centric system—to concentrate its financial resources on war activities. After the war, the government kept the wartime financial system to finance the reconstruction of the national economy. The government pursued so-called "policy finance," with commercial banks under generous protection and tight control. In the mid-1950s, a bank debenture market flourished. In 1966, the country issued bonds to finance capital investments for the first time after the war. The oil shock in 1973 triggered a severe recession in Japan, which from 1975 caused the government to issue bonds to finance current budget deficits. The Japanese government's serious efforts to develop its debt market began at this time.

Bond market reforms started in the mid-70s, with corporate bond market reforms beginning in the 1980s and being substantially completed in the late 1990s. The impact of the 1973 oil shock led in 1974 to the first year of negative growth in the postwar period. To stimulate the economy, the government ran a large budgetary deficit, which was financed by the issuance of Japanese Government Bonds (JGBs). The government initially placed JGBs with commercial banks which had a large pool of private savings in deposit accounts. The demand side of JGBs was also favorable; the economic slowdown and growing balance of payments surplus generated increased private savings. The savings fostered institutional investors such as insurance companies, trust banks, and investment trusts. They in turn sought better investment opportunities in JGBs. At the end of 2018, bond yields in Japan were close to zero percent, despite a debt-to-GDP ratio of the order of 250 percent.

5. Developing Domestic Government Bond Markets

A. Emerging market experience

The development of local currency bond markets (LCBMs) has become a matter of growing policy interest in EMs. Faced with growing budgetary deficits, several factors have forced governments to finance their deficit by tapping funds from their domestic markets. They include the limitations of banking

sector financing, inadequate availability of foreign aid and concessional foreign loans from the official sector (i.e., foreign governments and multilateral institutions), and increasing awareness of the risks associated with borrowing in foreign currencies.

EM LCBMs have grown over time, with a significant issuance of domestic debt across a range of economies. Some recent trends are summarized in Case Study 5.4; cross-country comparisons are discussed more fully in Chapter 2.

Case Study 5.4 LCBMs: recent trends

At the end of 2017, EM total debt stood at an estimated US$21.9 trillion of which general government accounted for an estimated US$10.3 trillion. The share of local currency debt rose to an estimated 87.1 percent in 2017.

Geographically, LCBMs have remained largest in the Asia Pacific region. From 2016 to 2017, government local currency debt outstanding in the region grew by US$1.1 trillion, mainly driven by increases in China and South Korea. However, there have been significant pockets of growth in LCBMs in several other EMs, including Brazil, Mexico, Slovakia, and South Africa.

The deepening of LCBMs is an essential tool for reducing foreign currency risk. That is not, however, a panacea since non-resident investors maintain large positions in several EM government bond markets, with substantial holdings of local currency bonds; and they feel exposed by a local currency depreciation. For example, in 2017 the share of non-resident investors in local currency government bond markets was above 30 percent in Indonesia, Mexico, Peru, Poland, and South Africa. China and India, two of the largest issuers of local currency bonds, have limited participation by non-residents, reflecting restrictions to access in those markets. Some countries have imposed capital controls to reduce the risk of sudden outflows but that itself brings risks if they damage investor perceptions of the underlying macroeconomic fundamentals. Participation of foreign investors in local currency non-government debt remains small, but is growing as domestic markets develop and an increasing number of instruments in EM currencies are issued in international capital markets.

The LCBMs of several emerging and low-income economies are now benefiting from attracting non-resident investors and extending maturities. In Africa, for example, Cote d'Ivoire, Namibia, Senegal, and Uganda more

than doubled the issuance of local currency government bonds between 2009 and 2014, with the stock of local currency bonds in these countries now on average equivalent to 8.5 percent of GDP. The maturity of bonds issued between 2009 and 2014 rose on average from 1.5 years to 6.4 years, with some countries such as Ghana, Kenya, Namibia, Nigeria, and Tanzania issuing local currency bonds in maturities over fifteen years.

The benefits of a vibrant domestic debt market go beyond providing a reliable source of financing for government deficits, and include many other positive externalities. The pursuit of developing countries in building deep and liquid markets also stems from the positive spin-offs they have on the development of the financial sector, its efficacy, and its flexibility in terms of monetary policy conduct and resilience to financial stability. Well-regulated, predictable, reliable, and liquid domestic debt markets can play a critical role in supporting economic growth, particularly in developing countries, at the macroeconomic and microeconomic levels.

The benefits of issuing longer-term Treasury bonds (T-bonds) are clear. By comparison with shorter-term securities, they minimize refinancing risk in the government debt portfolio and, by lengthening the average time to interest rate resetting, its exposure to interest rate risk. As the secondary market develops, it is the market prices of longer-term bonds that are the basis of the yield curve, against which corporate bonds can be priced and market risk hedged.

Establishing and developing domestic debt markets is a long and complex process that requires certain key preconditions. Many factors can inhibit the development of the market, including macro or political instability; financial controls; low domestic savings; paucity of institutional investors; proliferation of government agencies issuing securities, fragmenting the market; unpredictable issuance policy; and absence of the required market infrastructure. Potential obstacles to the development of a domestic market depend, therefore, on a country's overall degree and stage of development. Accordingly, to build a deep and liquid bond market, each country must develop its own reform plan suited to its conditions.

Debt managers are not responsible for ensuring that the macroeconomic preconditions are met. Instead, an inter-agency consultative process is required to establish the preconditions within the scope of a plan for overall macroeconomic reforms. Indeed, market development is probably best thought of as a project with many stakeholders and potentially with a

build-time of several years. There are set-up costs; and interdependences, prioritization, and sequencing all need to be addressed. The central bank and market regulators will need to be closely engaged and the private sector consulted or more directly involved. Policy making needs to be credible and consistent. Experience has shown that interventions are effective and reforms are best-enacted in countries where commitment begins with the top leadership and is conveyed to the ministerial level, particularly when key ministers are enlisted as partners to champion and implement reforms.

Essential preconditions

Initially, the government's main concern is to obtain the needed funding; after this, it can begin to focus on minimizing the funding cost and risks. Minimizing risks requires issuing longer maturities (reducing refinancing risk) at a fixed rate (reducing interest rate risk), and minimizing cost requires increasing the breadth and depth of the secondary market. Both objectives are linked. Investors are willing to buy longer maturities only if they are confident in their ability to sell the securities if they need cash. Investors are also willing to pay a higher price for a security with this advantage. In this case, the cost of funding the government is lowered, because the yield of a security declines when its price rises.

Credibility of the government as issuer of securities and rational policymaker

The political environment should be secure and the government should be credible before it issues securities. The legal framework should clarify the authority to borrow and issue new public debt, and to undertake transactions on the government's behalf. Investors and dealers need assurance that the debt office has legal authority to represent the government and that the government will stand behind the transactions into which it has entered. With this assurance, the market will focus on more advanced issues, such as whether the law adequately protects investors' rights; whether the regulatory environment ensures the safety of securities transactions; whether dependable legal procedures for dispute resolution provide for fair treatment; and whether the tax system is fair.

Uncertainty about future macroeconomic conditions—particularly about the course of inflation—will prevent the government from extending the yield curve beyond very short-term securities. If inflation is rapidly increasing and interest rates are high and volatile, investors will at best buy only very short-term securities with maturities no longer than a few weeks. High inflation

and high interest rates are indicators of economic and/or political problems, just like fever is an indicator of an underlying illness. Extension of the yield curve under persistent inflationary conditions may require issuance of inflation-indexed bonds or variable-rate bonds in the initial stage. A market can begin with a relatively high inflation rate; but to develop it needs government commitment to contain inflation.

Commitment of the government to pay market interest rates

The market cannot develop if the government enacts regulations to create a captive investor base by compelling some institutions to buy debt instruments (e.g., by obliging banks or pension funds to invest a certain percentage of their deposits in government debt), thereby enabling the government to issue at artificially low rates. Similarly, the market cannot develop if the government issues smaller amounts than announced or cancels a scheduled auction because of its subjective perception that asked yields are too high. The yields applicable to market instruments should be market-determined, not administratively set. The government should be committed to developing the market, financing itself through the market (not through captive investors), accepting market rates, and not manipulating auctions. The topic of "financial repression" is discussed at length in Chapter 6.

Guidance to the debt manager and transparency to stakeholders

The debt management strategy (MTDS) guides the government's financing choices, set out in the annual borrowing plan (ABP); and for all but the poorest or most fragile countries, a key component of the ABP will be the issuance of domestic securities. The targets of the MTDS for the main portfolio risk indicators are an important guide to developing the issuance plan, that is, the mix, size, and timing of the securities to be issued to meet the gross borrowing requirement implied by the annual budget. Conversely, market constraints on the design of the issuance plan will inform the periodic review and update of the MTDS.

The debt manager as issuer of government securities must therefore juggle many variables:

- objectives for the liability portfolio as expressed in the MTDS;
- the need to meet the government's financing requirement, taking account also of its profile across the year;
- the trade-offs between cost and risk implied by different instrument choices interacting with the trade-off expressed in the MTDS;

- the structure of demand, and investors' preferences as evidenced by the yield curve; and
- the importance of developing the domestic market.

Central to these decisions is building liquidity in government securities. The issuer benefits from greater investor demand and potential cost-savings. Investors benefit from reduced risk, and the ability to build a portfolio with the desired cost-risk characteristics. The wider market benefits from the great transparency of prices and yields, and the associated yield curve that is essential to pricing and the hedging of market risk.

Developing secondary market liquidity

Building liquidity has often proved a challenge. Many domestic government debt markets in EMs have grown impressively; but performance in the primary market has greatly outstripped that in the secondary market, which has often remained illiquid, with low turnover and little price transparency. Liquidity often suffers from a narrow range of investors and too many small (and therefore illiquid) bonds, which fragment the market. Even where the government is able to issue long-maturity bonds, they are often held by long-term saving institutions that are interested only in holding the bond to maturity to match known liabilities. The result is that EM governments pay a premium to investors for the risk of their holding relatively illiquid securities.

Not all these factors are outside the control of the debt manager. The need for coordination between the different agencies involved has already been stressed. But the debt manager has a direct role through the design of the issuance plan, and of the primary market. Central to this is the issuance of benchmark securities: large lines of T-bonds at key tenors. The large size improves the potential for wider distribution of the security among different types of investors with different incentives to trade, thereby increasing trading opportunities; it reduces search costs for those who want to buy; and it makes it less likely that any one transaction will change the market price. All these characteristics reduce the liquidity premium demanded by the market. EM government issuers are increasingly adopting a benchmark issuance policy to build sufficiently large lines of T-bonds as a necessary first step to foster secondary market activity. That in turn usually requires being able to issue successive tranches of the same bond; and it brings into relief the ability to deploy liability management operations. Such operations, whether buybacks or bond exchanges, can be used to build up a liquid bond of the chosen tenor;

and also manage cash management challenges as the larger bonds mature (see Section 6.C).

Predictability and transparency are the other essential characteristics of the issuance plan. It should be published—typically as part of the ABP with other documents supporting the annual budget. Investors will in turn be able to plan their own portfolios; and intermediaries their marketing strategies. Once it is published the issuer should try and keep to the plan—otherwise the benefits of publishing it are vitiated. Plans sometimes have to be changed, not least if the government's borrowing requirement changes in the course of the year. But it should be done in a way that is predictable—in the sense of consistency with previous policies. Indeed, "predictability" should be interpreted as predictability of policy responses: markets do not like surprises and the debt manager's decisions should be anchored back to well-defined and transparent objectives. Some countries also indicate at the start of the year the direction of any changes that might be necessary during the year.

Well-functioning money market

The money market is the cornerstone of a competitive and efficient system of market-based financial intermediation and plays several important roles. It facilitates monetary policy operations, with market-based instruments anchoring the short end of the yield curve and supporting the development of the foreign exchange market. It stimulates an active secondary bond market by reducing the liquidity risk attached to bonds and other term financial instruments, and by assisting financial intermediaries in managing liquidity risk. This latter aspect is especially important in supporting the development of primary dealers or market makers in government bonds: liquid short-term instruments are essential for financing their holdings of government bonds, which in turn underpins their functions of warehousing bonds—and reducing government's execution risk—in the primary market, and market-making in the secondary market.

Diversified investor base

A large and heterogeneous investor base with different risk preferences, investment maturity horizons, and trading motives ensures a strong and stable demand for government debt securities in a range of market conditions. It also gives more depth and liquidity to the market. This is not a precondition for a domestic debt market at inception; at the initial phase, only banking institutions participate. But, at a minimum, the country should have a sound

banking system that provides adequate appetite to invest in securities. Subsequently, contractual savings vehicles (institutional investors) such as pension funds, insurance companies, and mutual funds will provide a natural market for medium- and longer-term government debt.

Sound banking system

The soundness of the banking system is also relevant. Domestic and foreign investor concerns about the banking system will adversely affect the government's ability to roll over or issue new debt. It is essential that the banking system be subject to prudential regulations (including capital adequacy, lending standards, proper asset classification, income recognition, and reserving policies) that meet or approach international standards and provide for competent supervision and adequate enforcement capacity. At another level, lack of financially healthy intermediaries will damage secondary market liquidity. A banking system in crisis will further complicate government securities market development because important related markets, such as those for interbank and repurchase transactions, are unlikely to function properly.

Appropriate technical and regulatory infrastructure

In its initial phase, a securities market is merely a primary market: establishing the market basically requires only designing rules for auctioning securities and putting in place an elementary technical organization (e.g., a registry to give a legal title to instrument holdings, a central depositary for the custody of instruments, and a clearing and settlement system linked to a cash payment system so that instruments can be transferred). At inception, few transactions will be done in the market, and they will all be handled by banks that are assumed to be sophisticated investors. Thus, there is no need for a sophisticated, high-capacity technical infrastructure (e.g., delivery versus payment system) or detailed regulations protecting non-bank participants. As transaction numbers increase and the number of market participants diversifies, a more efficient system for the registration, custody, clearance, settlement of, and payment for debt instruments must be established to support further market development. The systems used to settle and clear transactions must be cost-efficient and easy to use. They should offer delivery vs. payment, and final registration of ownership. They will need to have a clear legal basis, be subject to regulatory oversight, and have the capacity to process required trading volumes within the chosen settlement cycle.

6. The Institutional Framework for Debt Management

A. Organization and governance

The emphasis on predictability, transparency, and credibility brings into focus the institutional framework for debt management. It requires legal clarity, a strong governance framework and well-specified organizational structures. The management of operations requires sound business practices (including operational risk management) with staff subject to a code of conduct and conflict of interest guidelines.

The governance structure has a number of elements, but important among them are:

- A process that separates high level policymaking from execution. This promotes transparency and accountability, and ensures that debt management policy and strategic portfolio objectives are properly embedded in longer-term macroeconomic objectives.
- Flowing from this strategy, a (published) mandate should be given to the debt managers—that is, the ABP, discussed in Section 5.A. The mandate is usually agreed by the minister or cabinet, together with structured delegated authorities to operate within the mandate subject to specified parameters.
- The governance framework should also cover decision-making processes and responsibilities, performance reporting, and internal and external audit.

The need for an organizational structure that supports professionalism, accountability, and focus on objectives, with some protection from day to day political pressures, is behind the international trend towards setting up a debt management function with a degree of operational independence. A separate office helps to establish credibility with the market; encourages a more market-responsive approach; and, depending on the civil service pay system, may make it easier to recruit and retain skilled staff. But there are disadvantages: separation puts more strain on the governance framework, with exposure to principal–agent risk. The greater the institutional separation, the more definition is needed in target setting and performance monitoring; that in turn requires skilled staff in the ministry as well as the debt office. Location closer to the ministry supports flexible interaction with the budget and planning functions, or makes it easier for other officials to make use of the debt office's

specific expertise. These different considerations, in particular principal–agent problems and coordination requirements, mean that, in emerging markets, it is usually advisable to establish a specialized function in or institutionally close to the ministry.

In practice there is a spectrum of institutional arrangements. In many countries, particularly in EMs, the debt management function is just another part of the treasury or ministry. But it may have separate "bureau" status within the ministry, sometimes as part of a wider treasury (New Zealand, Finland, United States, Brazil, Philippines, South Africa, Turkey); or be established as a semi-independent agency within the ministry (Australia, the Netherlands, UK, Belgium, France), or sometimes the central bank (Denmark, Iceland). There are a few cases where it has been developed as a semi-autonomous agency within government but not formally part of the ministry of finance (Sweden, Portugal, Ireland, Nigeria) or even as a company owned by government (Germany, Austria, Hungary), often to allow greater pay flexibility. Many of these offices still use the central bank as fiscal agent; all use the bank for some services. Several, particularly in Europe, have important cash management functions, discussed below, and may also have responsibilities for, for example, management of assets or contingent liabilities.

B. The interplay between cash and debt management in safeguarding access to liquidity

The overriding objective of cash management is to ensure that the government is able to fund its expenditures in a timely manner and meet its obligations as they fall due. However, cost-effectiveness, risk reduction, and efficiency are also important, and specifically:

- minimizing the costs of holding cash balances in the banking system;
- reducing risk: operational, credit, and market risk;
- adding flexibility to the ways in which the timing of government cash inflows and outflows can be matched;
- supporting other financial policies, in particular debt management policy, monetary policy, and the development of domestic financial markets.

There are distinct phases in the development of a modern cash management function:

- Developing the Treasury Single Account (TSA)—the integration of government bank accounts, and the sweeping of overnight balances into a single account or a network of linked accounts held by the Treasury at the central bank.
- Building a cash flow forecasting capability—the development of capacity within the Treasury to monitor and forecast flows in and out of government—that is, changes in the balances in the TSA.
- Moving to more active cash management—borrowing and lending in the money market to a pattern deliberately designed to smooth or reduce the volatility of net daily cash flows.

T-bills are the usual instrument of choice in moving to more active cash management. Net T-bill issuance will be higher or lower in any week depending on whether outflows are expected to be higher or lower than inflows in that week. The forecast should also guide the maturities of the securities to be issued, as well as the volumes (and potentially also the volumes and maturities of any investments of temporary cash surpluses), with a view to smoothing the cash flow across the TSA. A smoother cash flow means lower average cash balances with reduced net borrowing costs and also less pressure on the central bank's monetary policy operations (because, other things equal, the mirror image of fluctuations in the TSA is fluctuations in banking sector liquidity).

T-bills are a core instrument in domestic financial markets. As a risk-free instrument they are important to banks, to meet risk and liquidity requirements; to asset managers, to facilitate achieving the chosen cost–risk tradeoff; and to the authorities, to meet debt management, cash management, and monetary policy objectives.[1] They are also in demand as collateral. Government cash management is focused on a much shorter time period than debt management. Short-term T-bills are more useful than T-bonds or longer-term bills. It may be possible to issue T-bonds to refinance maturing T-bonds; but the underlying primary balance will have a profile linked, for example, to the quarterly or monthly timing of tax receipts, salaries, or transfers. Many countries use one-month T-bills for cash management, or T-bills with odd maturities linked to days of future inflow; the United States uses two-week T-bills. The volume issued can be more readily varied to offset peaks and troughs in the cash profile. T-bills with a maturity of three months or more

[1] T-bills can be used as monetary policy instrument by the central bank requesting the ministry to issue additional T-bills, the proceeds of which are then sterilized by being held in an account at the central bank. This technique potentially has some advantage over the central bank issuing its own bills, which may fragment liquidity in the nascent bill market with two very similar instruments competing.

are less flexible and the stock outstanding is more often held steady in line with investors' demand and portfolio requirements.

Repo[2] is the instrument of choice for fine tuning or for borrowing and lending outside the normal T-bill issuance schedule. Repo has the great advantage that the lending is collateralized, reducing any credit risk concerns. It is also very flexible, in both the speed of execution and the range of maturities available. Many settlement systems are able to settle transactions on the same day, also handling the collateral automatically. For the same reasons, repo has a central role in the development of the financial market.

Central banks rely heavily on repo for liquidity and short-term interest rate management. It can stimulate the interbank market by removing credit risk concerns. Primary dealers can repo out securities to finance purchases of the same securities. Repo can support short-selling, another characteristic of efficient market making, with the security sold repoed in ahead of settlement. More active cash management will itself be a stimulant, including of the T-bond market, since domestic T-bonds are normally the preferred collateral. These linkages are illustrated in Figure 5.4. This benefits the debt

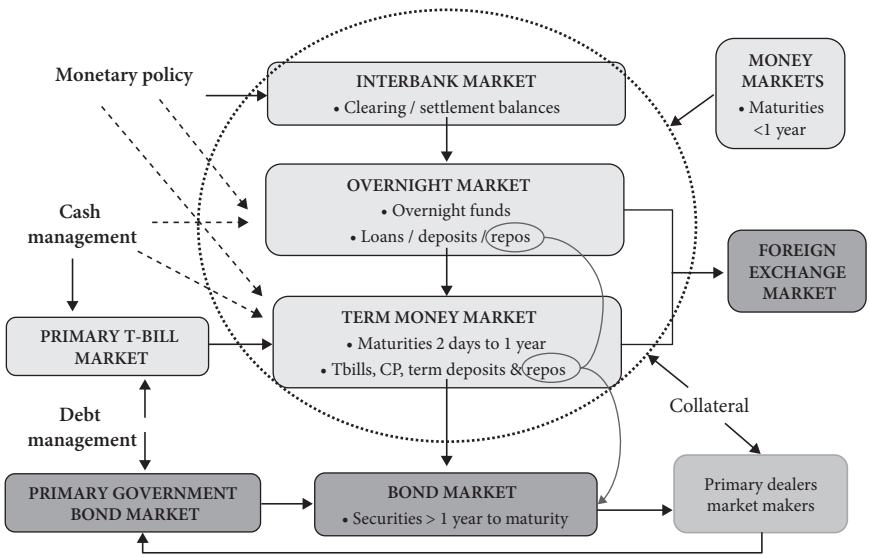

Figure 5.4 Money *Market: Interaction with other Financial Markets*
Source: Modified from: Mike Williams (2010).

[2] A "repo" (short for sale and repurchase agreement) is the sale of securities tied to an agreement to buy them back later. A reverse-repo is the purchase of securities tied to an agreement to sell back later. A repo is best thought of as a collateralized loan. T-bills or T-bonds (or central bank paper) are the dominant collateral for repo transactions, particularly in the early development of the market.

manager who, as well as issuing T-bonds, may also need to lend or repo bonds temporarily to the market makers to ensure that they are always able to provide a two-way market.

C. Importance of cash and debt management coordination

Debt and cash managers must work closely together; indeed, in many countries the functions are now integrated, whether in a fully-fledged debt management office, of the kind in many eurozone countries, the United Kingdom, Sweden, Hungary, and Australia; or through merger or integration between treasury and debt management directorates, as in Peru, Colombia, Turkey, and South Africa.

Debt managers have to juggle the full range of instruments in making decisions about issuance. They have to trade off from day to day, week to week, and month to month, the demands of the strategy and the demands of the market, with issuance choices, of both T-bills and T-bonds, taking into account demand, supply, and price information. In relation to demand, intermediaries and/or end-investors may need a steady flow of T-Bonds to meet their obligations, or shorter-term instruments for liquidity management. Their needs will change across the year with their own cash flows and market developments. From the supply perspective, government financing choices have to be made in the context of the profile of financing flows, whether it reflects the profile of the primary deficit or of debt servicing payments.

There are other day-to-day coordination requirements between debt and cash managers including:

- Linkage of issuance dates with redemption dates, to maximize the opportunities for investors to roll over into a new issue.
- Maturity dates should also be chosen to avoid weeks, and especially days, of heavy cash outflow (e.g., salary payments); and indeed, should target days of cash inflow (the due date for tax payments).
- Debt managers can, through liability management operations, mitigate the cash management problems that potentially arise when large bonds come to maturity.

The costs of a failure to coordinate debt and cash management decisions is not immediately apparent. Some of it arises from the income forgone from investing surplus cash; and since debt servicing payments are usually given priority in the parliamentary approval processes, the costs of extra borrowing

are not subject to the same scrutiny as other expenditures. But there is a cost. This is apparent in those EMs who in recent years have very properly built their debt management capacity, but left the longer-established treasury function relatively untouched. Since debt managers tend to focus on the redemption profile, it is left to the treasury to cope with in-year fluctuations; but the only tool that they have is building an expensive cash buffer—the interest earned on surplus cash is much less than the cost of additional borrowing. For prudential reasons, some countries also frontload debt issuance to build a cash buffer, which can similarly be costly. At least some countries are now alive to the benefits of investing temporary surpluses back into the banking system; both to earn more interest and to help smooth cash flows: China and Russia are examples among larger countries.

There is widespread awareness of the importance of developing the TSA, and even the poorest countries have taken some steps to do so, and are starting to build a forecasting capability. The move to more active cash management may take longer, and has to be linked to the development of the money market. But the potential strain between debt and cash management objectives over whether to issue T-bonds or T-bills when faced with an imminent cash shortage is lessened as the scope for active cash management develops. The issuance of bonds with a stable and predictable pattern reduces market uncertainty and intermediaries and investors can better plan ahead. With a liquid money market, the timing of bond sales can be separated from the profile of the government's net cash flow. It is left to T-bills and other money market instruments to deal with the short-term fluctuations. That in turn greatly improves the transparency and efficiency of debt management.

As interaction with the market develops, the integration of debt and cash management functions becomes especially important. It ensures that the government presents a consistent face to the market. Where two parts of government are interacting with the market, there are risks of giving conflicting signals, adding to uncertainty, and also potentially distorting the money market. Front office staff directly managing the transactions in the market may also need to intervene in the money market for debt management reasons, or in the debt market for cash management reasons.

D. Role of debt as a safe asset

Debt plays an important role as a safe asset and has policy relevance on several levels: it (i) provides a benchmark yield curve for the corporate debt

market; (ii) supports liquidity management operations of the central bank; (iii) provides an investment alternative with little or no risk of default for investors; (iv) maintains and develops smoothly functioning financial markets; and (v) provides market infrastructure through a robust payment and settlement system and a strong legal framework (i.e., collateral and bankruptcy laws).

Safe assets are used as a reliable store of value and aid capital preservation in portfolio construction. They are a key source of liquid, stable collateral in private and central bank repo agreements and in derivatives markets, acting as the "lubricant" in financial transactions. As key components of prudential regulations, safe assets provide banks with a mechanism for enhancing their capital and liquidity buffers. As benchmarks, safe assets support the pricing of other riskier assets. Finally, safe assets have been a critical component of monetary policy operations.

These widely varying roles of safe assets and the differential price effects across markets make it difficult to gauge the overall price of safety. A tightening of the market for safe assets can have considerable implications for global financial stability, including an uneven or disruptive pricing process for safety. As investors scramble to attain scarce safe assets, they may be compelled to move down the safety scale, prompting the average investor to settle for assets that embed higher risks.

Implementation of the Basel III capital and liquidity framework also has important implications. The weighing of assets and risk weighing of government securities will have an impact on the level of capital adequacy. The Basel III liquidity framework, which defines high quality liquid assets, may not fully reflect financial market structures in EMs, impacting the functioning of domestic financial markets and raising the need for national discretion to, for example, adjust for lack of market liquidity in the relevant assets.

In response to the global financial crisis, authorities in many jurisdictions are encouraging greater use of central counterparties (CCPs) for over-the-counter (OTC) derivatives transactions. In particular, the G20 has agreed that all standardized OTC derivatives should be centrally cleared so as to lower counterparty credit risk through multilateral netting. The global nature of OTC derivatives markets has also highlighted the need for international coordination to establish minimum cross-border risk management standards and avert regulatory arbitrage in cases where CCPs compete with each other. The expected changes in OTC market infrastructure will likely increase demand for safe assets via higher demand for collateral.

A shift toward central clearing of standardized OTC contracts will eliminate some of the need for bilateral collateralization; but the move of a critical

mass of OTC derivatives to CCPs is expected to increase the demand for collateral. The higher demand would arise from an initial margin that typically is not posted on bilateral interdealer trades, and from contributions to guarantee funds at the CCP. Moreover, a proliferation of CCPs without mutual recognition may raise total collateral requirements even further. The lower estimate is associated with exemptions of certain types of OTC derivative counterparties (such as sovereigns and "hedgers") or types of contracts (such as foreign exchange derivatives) from the central clearing mandate. More importantly, restrictions on the market reuse (rehypothecation) of collateral posted with CCPs may lower the effective supply of collateral in the market and hence increase the liquidity risk premium (Singh 2011).

7. Concluding Remarks

We conclude by presenting some experiences and considerations with regard to issues faced by debt managers. In their consideration of the risks and trade-offs when developing financing strategies and issuance plans, a distinction may need to be made between developed and developing economies. Middle- and low-income economies may lack market access and, as a result, are often forced to accept additional risks and costs. As noted above, for short-term domestic debt, governments face an intertemporal trade-off, that is, excessive reliance on short-term or floating rate debt to take advantage of lower short-term rates. However, access to long-term financing is sometimes simply not available (at any cost). Low-income countries need to build their yield curve and lengthen maturities; but short-term external debt might be the only option for those that do not have sufficiently robust domestic markets or domestic financing sources. Issuing short-term debt can pose a significant refinancing risk, especially for infrequent issuers or those with macroeconomic imbalances, where market conditions can change radically, bringing exposure to sudden stops.

Debt managers are often preoccupied with the issuance of sovereign guarantees and accounting for related contingent liabilities. Many have expressed concern that they have imperfect visibility of debts contracted by state-owned enterprises or other government agencies, despite their potentially large impact on borrowing limits. These debts are often not reported to the debt management office (or Ministry of Finance) and are therefore difficult to monitor. Adjustments to the accounting of contingent liabilities based on credit analysis may be made in developed economies, but not always applied elsewhere.

Finally, it should be noted that many preconditions for developing robust capital markets are out of the control of the debt manager, including (i) political stability and government credibility; (ii) macroeconomic stability and confidence about the course of inflation; (iii) prudent fiscal policy; (iv) financial controls; (v) domestic savings rates (and more broadly the depth of financial inclusion); (vi) domestic institutional investor base; (vii) government discipline over the issuance of debt (and sovereign guarantees); and (viii) functioning technical and regulatory infrastructure. However, debt managers can have a profound impact on the development of financial markets by promoting a strong sense of predictability and transparency to stakeholders.

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